AN2DL - Second Challenge Report
Team Name: How I Met Your Model

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1 Introduction

Microscopic tissue analysis plays a key role in cancer diagnosis and treatment planning, yet clinical practice still relies largely on manual inspection of whole-slide images (WSIs). This challenge addresses automatic classification of molecular subtypes from low-magnification histopathology patches extracted from WSIs. Each image is classified into one of four breast cancer subtypes—Luminal A, Luminal B, HER2-positive, and Triple Negative—using only hematoxylin–eosin-like morphology. Evaluation is based on the macro F1-score, which accounts for class imbalance and emphasizes performance on difficult subtypes such as Triple Negative.

Rather than performing whole-image classification, we model each slide as a bag of mask-guided patches. Multiple high-resolution crops are sampled around mask components and used to train a convolutional model for slide-level subtype prediction. At inference, predictions from multiple patches are aggregated to produce a robust slide-level estimate.

2 Problem Analysis

The dataset comprises 691 training and 477 test images, each accompanied by a binary tumor mask. The task is challenged by high variability in image resolution and visual quality, where slides are frequently dominated by background or benign tissue with spatially sparse masks. Additionally, label noise is introduced by artifacts such as cartoon-like "Shrek" slides. Class imbalance is significant, particularly for Triple Negative (11.14%, $n = 77$) compared to Luminal B (31.84%), Luminal A (29.67%), and HER2+ (27.35%)4. These factors necessitate a weakly supervised MIL approach to effectively aggregate localized discriminative information5555.

3 Method

Our approach comprises four components: (i) mask-guided patch preprocessing, (ii) a dual-path architecture that encodes both RGB tissue appearance and mask geometry, (iii) a class-aware training strategy with strong regularization, and (iv) a patch aggregation scheme for slide-level prediction. The overall design is optimized to maximize slide-level macro F1.

3.1 Preprocessing & Patch Extraction

Before patch extraction, a data cleaning step was applied to address artifacts identified during problem analysis. Cartoon-like "Shrek" slides were removed entirely. Slides containing green marker stains were retained and processed using an automated removal procedure. Specifically, green pigment was detected in the HSV color space (Hue 30–90), refined via morphological dilation, and the affected regions were inpainted using the estimated background color. Corresponding areas in the ground-truth masks were zeroed to prevent spurious supervision.

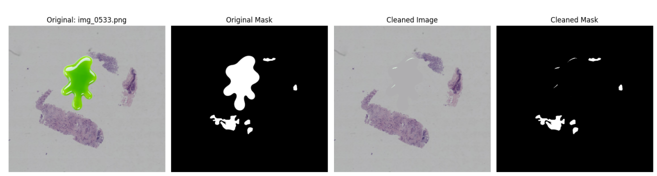

Figure 1: Comparison of a stained sample before and after preprocessing.

Given the heterogeneity of WSIs and the sparsity of informative tissue, fixed-size patches are extracted from each image–mask pair. A sliding window of 224×224 pixels with stride 95 (approximately 50% overlap) is used, retaining only patches with mask coverage greater than 5%. Outlier patches are removed using brightness and variance thresholds to filter overly dark, bright, or low-texture regions, and near-duplicate patches within each slide are eliminated using cosine similarity > 0.98. RGB patches are normalized to $[0, 1]$, with corresponding mask crops scaled accordingly, yielding a compact multi-patch representation per slide.

Mask-conditioned augmentation.

A lightweight texture augmentation is applied exclusively within masked regions, introducing local stain-intensity shifts, Gaussian-smoothed micro-texture noise, and contrast perturbations, while preserving pixels outside the mask. This encourages the model to focus on morphological structure rather than superficial staining variations.

Each patch inherits its slide-level label, resulting in a weakly supervised multi-instance dataset. To prevent information leakage, train/validation splits are performed at the *slide* level, with all corresponding patches following the assignment. The final dataset consists of normalized RGB patches, normalized binary mask patches, slide identifiers, and labels where available, enabling patch-level training with slide-level aggregation during evaluation.

3.2 Dual-Path Architecture

We employ a dual-path architecture that integrates morphological and geometric cues [1]. The RGB branch uses a ViT-Small backbone pretrained with DINO on histopathology data [2], producing a global embedding via token pooling. In parallel, the mask branch is a lightweight convolutional encoder composed of stacked Conv–BN–ReLU layers with downsampling, generating a 256-dimensional representation of lesion shape and spatial distribution. The RGB and mask embeddings are concatenated and passed through a small multilayer perceptron with dropout to produce four-class predictions.

3.3 Training Procedure

The model is trained end-to-end using cross-entropy loss with inverse-frequency class weighting to emphasize HER2-positive and Triple Negative samples, combined with label smoothing (0.05). The RGB backbone is fine-tuned with a learning rate of 1×10^{-4}, while the mask and fusion layers use 1×10^{-3}. Standard data augmentations, including flips, rotations, and color jitter, are applied to RGB patches. Optimization is performed using AdamW with weight decay and mixed-precision training. Early stopping is based on *slide-level* macro F1, computed by aggregating predictions across all patches from the same slide.

3.4 Slide-Level Inference

During inference, all patches from a slide are independently processed by the dual-path model, and their softmax outputs are averaged to obtain a slide-level probability vector used for subtype prediction. This aggregation reduces patch-level noise and aligns with the challenge evaluation protocol. Test-time augmentation using flips and rotations is applied to further improve robustness.

4 Experiments

All models are trained on the patch–mask dataset using identical 80–20 train/validation splits, with evaluation performed via slide-level aggregation. Performance is reported as slide-level macro F1 on the internal validation set, while final results correspond to the official Kaggle leaderboard metric. We evaluate progressively more expressive variants, including baseline CNNs, dual-path mask-aware models, contrastive pretraining, generative augmentation, post-hoc modeling, and refinement strategies such as pseudo-labeling and patch selection.

4.1 Model Comparisons

Table 1 summarizes the performance of all evaluated configurations. A patch-based ResNet50 establishes a baseline of 37.4% locally and 37.8% on the leaderboard. EfficientNet-B2 shows lower local performance but improved external robustness. Incorporating a dual-path design with EfficientNet-B2 improves both metrics, with a further modest local gain from mask-based generative augmentation. The best single-model performance is achieved by the dual-path Lunit DINO ViT-S backbone with mask-conditioned augmentation, reaching 41.98% local F1 and 41.18% on the leaderboard. Pseudo-labeling consistently reduced performance, while a selective "k-best" patch strategy produced competitive but inferior results.

We further evaluated an ensemble strategy based on prediction diversity. Using submission commonness analysis (Figure 2), we selected the four highest-performing and four most diverse models. Aggregating their predictions produced a substantial improvement, yielding a final macro F1 of 43.71%.

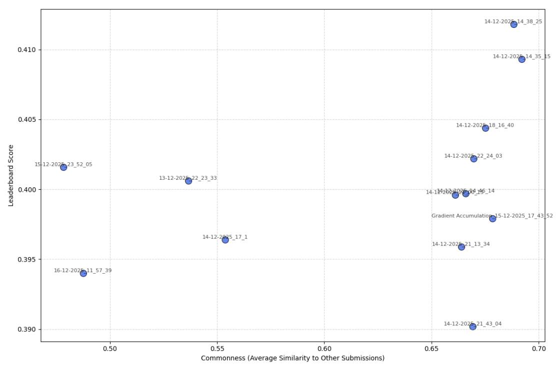

Figure 2: Analysis of Leaderboard Score vs. Prediction Commonness.

Table 1: Performance of the main model variants (macro F1, %). Best results are highlighted in **bold**.

Model Variant	**Local**	**Kaggle**
ResNet50 (patch-based)	37.44	37.79
EfficientNet-B2 (patch-based)	31.96	38.82
DP + EfficientNet-B2	39.98	40.06
DP + EfficientNet-B2 + mask gen.	40.01	38.75
DP + Lunit DINO + mask gen.	**41.98**	**41.18**
DP + Lunit DINO + mask gen. + pseudo-labeling	39.99	36.78
DP + Lunit DINO + mask gen. (k-best patches)	38.33	39.96
DP + Lunit DINO + mask gen. + XGBoost Head	40.77	40.44
SimCLR + DP + Lunit DINO + mask gen.	39.02	31.56
Ensemble (Best 4 + Diverse 4) **(18th in the leaderboard)**	-	**43.71**

4.2 Findings

The results show that (i) mask-aware patch selection is essential for stable learning, (ii) pathology-pretrained transformers (Lunit DINO) consistently outperform CNNs, and (iii) extensive patch aggregation is more effective than pseudo-labeling or selective patch filtering. The final system achieves over 41% macro F1 on both internal validation and the public leaderboard, ranking among the top-performing solutions.

5 Results

Prior to adopting a patch-based formulation, whole-image models plateaued at approximately 0.27 macro F1 on the leaderboard, indicating that global representations failed to capture the sparse and localized morphology required for molecular subtyping. Introducing mask-guided patch extraction led to immediate performance gains across all architectures, confirming the importance of fine spatial resolution and multi-instance evidence.

Overall, the results highlight three key observations: (i) patch-based learning is essential for extracting meaningful signal from low-magnification WSIs; (ii) explicit integration of mask information yields consistent improvements; and (iii) pretrained vision transformers provide the strongest representations for this domain.

6 Discussion

The results demonstrate that patch-based, mask-aware transformer models are well suited for capturing fine-grained morphology and leveraging lesion geometry in weakly supervised settings, but they also present limitations. Although the dual-path ViT-S architecture achieves the strongest performance, it relies on accurate mask extraction and assumes that masked regions consistently contain discriminative features, which may not hold for all slides. The approach remains sensitive to label noise from slide-level annotations, as evidenced by the degradation observed with pseudo-labeling. Moreover, while extensive patch aggregation improves robustness, it increases computational cost and assumes that simple averaging adequately reflects the true slide-level distribution.

7 Conclusions

We present a patch-based, mask-aware dual-path architecture for molecular subtype prediction from low-magnification histopathology images, achieving substantial gains over whole-image baselines. Key contributions include a robust patch extraction pipeline, explicit integration of lesion geometry via a mask encoder, and the use of pathology-pretrained transformers. Future work may investigate advanced multi-instance learning formulations, attention-based patch aggregation, and improved stain normalization or color-invariant representations.

References

[1] J. Li, S. Wu, and B. Zhang, "Multi-modal Dual-Path Attention Network for Medical Image Classification," IEEE Transactions on Medical Imaging, vol. 40, no. 5, pp. 1450-1460, 2021.

[2] C. Chen and R. Krishnan, "Histo-ViT: A Vision Transformer for Histopathology Whole Slide Images," arXiv preprint arXiv:2201.07662, 2022.